

Rushikesh Suresh Nikumbh

Mechanical Engineer | Clean Energy Technologies | R&D Validation

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PROFILE

Mechanical Engineer with an M.Sc. in Clean Energy Technologies from FAU Erlangen (thesis grade 1.8) and hands-on experience in **fuel cell stack testing, validation, prototype design, and manufacturing engineering**. Skilled in operating test benches for functional and performance testing, developing fixtures and tooling, and evaluating results through structured **DVP&R**. Comfortable learning new technical systems independently, from CNC programming to electrochemical test rigs, and troubleshooting methodically when something does not behave as expected. Brings together practical workshop experience, strong CAD and documentation skills, and a background spanning both industrial manufacturing and applied energy research.

CORE COMPETENCIES

Electrochemical Systems	PEM fuel cell stack components • Graphite bipolar plates (BPPs) • MEA bonding & sealing • Alkaline water electrolysis • Stack assembly • Polarisation curves / J-V analysis
Prototyping & Workshop	Test rig development • Fixture design • Vacuum tooling (vacuum gripper & vacuum table) • 3D printing (FDM/SLA) • CNC dispensing (NC/G-code) • Injection moulding • Prototype assembly • Workshop tools
Testing & Validation	Fuel cell stack testing • Functional / performance testing • Leak testing (150–2,300 mbar) • Fuji pressure test • Design verification plan & report (DVP&R) • Lap shear (ISO 5478) • Tensile (ISO 527-1) • Shore A hardness (ISO 868) • Accelerated lifetime testing (1,000 h / 95 °C) • Calibration checks • Test report writing
Measurement Technology	Profilometer • 3D scanning • Displacement measurement • Temperature measurement / thermocouples • Measurement data evaluation • Fault diagnosis • Test specimen installation
CAD & Documentation	SolidWorks • CATIA • PTC Creo • AutoCAD • MATLAB (basic) • Technical drawings • Test protocols • DVP&R reports • Work instructions • Supplier quotations • Hazardous materials safety docs

SELECTED ENGINEERING IMPACT

- **Designed a 12-station oven curing layout** for 342 BPPs/day (vs 20/day with a single station), defining process parameters, tooling concepts, acceptance criteria, material flow, and operator workforce requirements.
- **Converted DVP&R requirements into structured test procedures**, defining test conditions, acceptance criteria, sample preparation, measurement intervals, and reporting requirements while considering the actual fuel-cell operating environment (DI water and Glysantin, alongside temperature).
- **Developed a snap-fit BPP alignment tool** (3D-printed) that eliminated the need for 2 additional operators, and designed a new fixture to support bonding of a new BPP design concept for rapid prototyping.
- **Built and tested an alkaline electrolyser stack** independently, operating a DC power supply, monitoring voltage and current, and characterising performance across varying temperature and electrolyte conditions.

WORK EXPERIENCE

Working Student — Fuel Cell Test Fixture Development

11/2025 – 12/2025

Technical University Munich, Dept. of Chemistry

Munich, Germany

- Designed and 3D-printed a ceramic fuel cell single-cell test fixture with sealing, clamping, and electrochemical interface, and performed **leak-tightness verification** and electrochemical voltage characterisation.
- Produced CAD models, technical drawings, assembly instructions, and commissioning procedures.

Thesis Student — Fuel Cell Stack Testing

01/2025 – 06/2025

Freudenberg e-Power Systems

Munich, Germany

- Designed, built, and operated experimental test setups for bonding validation, integrating thermocouples for temperature measurement and process-data evaluation across IR heating, hot-plate, and oven curing, and selected the oven curing process based on test data and economic considerations.
- Developed a **dedicated leak-test fixture** for small BPP samples to evaluate sealing performance and leakage behaviour.
- Developed and executed **DVP&R test plans** covering leak testing, tensile, lap shear, Shore A hardness, and chemical stability, writing test protocols and test reports per test specifications.
- Learned the **DATRON CNC dispensing system's code structure independently** and wrote new programs for specific dispensing outputs, achieving a **6× throughput improvement** by optimising acceleration and speed factor parameters alongside tooling improvements.
- Designed and manufactured custom fixtures, vacuum tooling, alignment systems, and 3D-printed hardware for repeatable stack-component installation, assembly, and validation testing.
- Investigated failure mechanisms through structured testing and **root-cause analysis**, collaborating closely with R&D engineers, technicians, suppliers, and production teams to support testing, troubleshooting, and manufacturing improvements.

R&D Intern — Fuel Cell Stack Testing

08/2024 – 12/2024

Freudenberg e-Power Systems

Munich, Germany

- Performed displacement, leakage, and Fuji pressure testing, along with stack analysis and geometrical inspections using visual inspection, 3D scanning, and profilometer analysis to support process validation and quality evaluation.

- Supported hands-on workshop activities, prototype troubleshooting, and rapid engineering modifications in fast-paced development environments.
- Modified validation procedures to detect bonding defects earlier in the process, reducing rework and improving production workflow reliability.
- Optimized 3D printing processes by designing a customized base pattern that improved surface finish and straightness while reducing material consumption during prototype manufacturing.
- Produced technical documentation, including risk assessments, operating instructions, supplier quotations, and hazardous materials documentation.

Working Student — Bipolar Plate Materials Research

FAU Erlangen — Lehrstuhl für Kunststofftechnik (LKT)

03/2024 – 07/2024

Erlangen, Germany

- Developed graphite/PP composite bipolar plates (80% graphite / 20% PP) via injection moulding, gaining practical insight into material behaviour and manufacturing constraints.
- Researched **frame-integrated MEA bonding and sealing concepts** for scalable fuel cell stack assembly, and coordinated experimental work to document results for research outputs.

Graduate Trainee — Manufacturing Engineering

Truetech Vision Industry Pvt. Ltd.

07/2020 – 01/2021

India

- Gained hands-on experience in manufacturing machine operation, CAD-based tooling development, and process planning, building a foundational understanding of metal material behaviour for forged and cast component analysis.

Thesis Intern — Design Engineering

Pragati Pvt. Ltd. (ISO 9000 / TS16949 certified)

06/2019 – 04/2020

India

- Designed and manufactured a complete 2-cavity injection mould (MS/P20 steel) for a nylon wheel component, delivered at ₹18,453 and achieving **~80% cost reduction** versus the machined alternative.
- Identified trapped air as the root cause of injection defects, redesigned the runner and venting systems, and executed a 3-phase mould trial to reach a production-ready standard.

KEY PROJECTS

Low-CO₂ Scenario Modelling — Jena City | M.Sc. Mini-Project

- Modelled a city-scale energy transition integrating fuel cells, electrolysis, bioenergy, solar, EVs, and CHP, quantifying a 21.2% CO₂ reduction pathway.

Alkaline Water Electrolysis Prototype — Independent R&D Project

- Designed, built, and tested an alkaline electrolyser stack from scratch, characterising hydrogen production and degradation behaviour under varying temperature, electrolyte concentration, electrode geometry, and stack configuration.
- Operated a transformer-based DC power supply and used an ammeter and voltmeter to monitor and record stack current and voltage, performing electrical connections, polarity verification, and safe setup checks throughout testing.

Experimental Investigation of Connecting Rod — Undergraduate Project

- Performed UTM testing and material characterisation (Vickers hardness, microscopy, inclusion analysis) to evaluate toughness and failure behaviour.

EDUCATION

M.Sc. Clean Energy Technologies (Clean Energy Processes)

Friedrich-Alexander University Erlangen-Nürnberg (FAU)

10/2021 – 2026

GPA 2.5 | Thesis 1.8 (sehr gut)

Key modules: Fuel Cells & Electrolysers • Electrical Energy Storage • Polymer Science • Chemical Technologies for Energy Transition • Life Cycle Assessment

B.E. Mechanical Engineering

Savitribai Phule Pune University

08/2017 – 05/2020

CGPA 7.85 — First Class with Distinction

Key modules: Machine Design • FEA • Thermodynamics • Materials Science & Metallurgy • Electrical & Electronics Engineering

Diploma in Mechanical Engineering

MSBTE, India

08/2014 – 06/2017

82%

LEADERSHIP & COMMUNICATION

Event Head

Technical Event, Bachelor Studies

- Led the organisation of a large technical event, working directly with external sponsors to understand requirements, managing expectations as circumstances changed, and delivering on commitments across a professional boundary.

LANGUAGES & ADDITIONAL EXPERIENCE

English Business Fluent (C1)

German B1 (studying toward B2)

Marathi Native (C2)

Hindi Business Fluent (C1)

Shreyas Industry (Trainee) Process flow mapping, BOP documentation, ISO quality checks

Kalpataru Precision Tools (Intern) Surface finishing and dimensional accuracy in drilling, tapping, lathe operations